

List of Publications

1. Pavel Koupil, Jáchym Bártík, and Irena Holubová. “MM-evocat: A Tool for Modelling and Evolution Management of Multi-Model Data”. In: *Proceedings of CIKM 2022*. Atlanta, GA, USA: ACM, 2022, pp. 4892–4896. ISBN: 9781450392365. DOI: [10.1145/3511808.3557180](https://doi.org/10.1145/3511808.3557180) (Paper II)
2. Jakub Klímek et al. “Atlas: A Toolset for Efficient Model-Driven Data Exchange in Data Spaces”. In: *Proceedings of MODELS 2023*. IEEE. 2023, pp. 4–8. DOI: [10.1109/MODELS-C59198.2023.00009](https://doi.org/10.1109/MODELS-C59198.2023.00009)
3. Pavel Koupil, Jáchym Bártík, and Irena Holubová. “MM-evoque: Query Synchronisation in Multi-Model Databases”. In: *Proceedings of EDBT 2024*. OpenProceedings.org, 2024, pp. 818–821. DOI: [10.48786/edbt.2024.78](https://doi.org/10.48786/edbt.2024.78) (Paper III)
4. Jáchym Bártík. “AI-Powered Orchestration of Multi-Model Data”. In: *Proceedings of VLDB 2024 PhD Workshop*. Guangzhou, China: VLDB Endowment, Aug. 2024. URL: https://vldb.org/2024/files/phd-workshop-papers/vldb_phd_workshop_paper_id_2.pdf (Paper I)
5. Jáchym Bártík, Pavel Koupil, and Irena Holubová. “Modelling and Evolution Management of Multi-Model Data”. In: *Proceedings of SAC 2024*. ACM, 2024
6. Jáchym Bártík, Alžběta Šrůtková, and Irena Holubová. “TransforMMer: A Universal Multi-Model Data Generator”. In: *Proceedings of EDBT 2025*. OpenProceedings.org, 2025, pp. 1150–1153. DOI: [10.48786/edbt.2025.111](https://doi.org/10.48786/edbt.2025.111) (Paper IX)
7. Pavel Koupil, Jáchym Bártík, and Irena Holubová. “Model-Agnostic Evolution Management”. In: *Proceedings of RCIS 2025*. Vol. 547. LNCS. Springer, 2025, pp. 245–261. DOI: [10.1007/978-3-031-92474-3_15](https://doi.org/10.1007/978-3-031-92474-3_15) (Paper IV)
8. Irena Holubová, Alžběta Šrůtková, and Jáchym Bártík. “Reshaping Reality: Creating Multi-Model Data and Queries from Real-World Inputs”. In: *Proceedings of ENASE 2025*. Jan. 2025, pp. 174–184. DOI: [10.5220/0013395300003928](https://doi.org/10.5220/0013395300003928) (Paper VIII)
9. Filip Ježek et al. “DortDB: Bridging Query Languages for Multi-Model Data Ponds”. In: *Proceedings of VLDB 2025* 18.12 (Aug. 2025), pp. 5335–5338. ISSN: 2150-8097. DOI: [10.14778/3750601.3750665](https://doi.org/10.14778/3750601.3750665) (Paper X)
10. Pavel Koupil et al. “FDepHunter: Harnessing Negative Examples to Expose Fakes and Reveal Ghosts”. In: *Proceedings of VLDB 2025* 18.12 (Aug. 2025), pp. 5227–5230. ISSN: 2150-8097. DOI: [10.14778/3750601.3750612](https://doi.org/10.14778/3750601.3750612) (Paper V)
11. Filip Mihál, Jáchym Bártík, and Pavel Koupil. “Refining Storage Strategy through Index Selection Methods in Multi-Model Database Systems: A

Survey”. In: *Data & Knowledge Engineering* 164 (2026), p. 102582. ISSN: 0169-023X. DOI: [10.1016/j.datak.2026.102582](https://doi.org/10.1016/j.datak.2026.102582) (Paper VI)

12. Pavel Koupil et al. “MM-mapsearch: Workload-Aware Mapping Selection”. In: *Proceedings of EDBT 2026*. OpenProceedings.org, 2026. DOI: [10.48786/edbt.2026.68](https://doi.org/10.48786/edbt.2026.68) (Paper VII)
13. Jáchym Bártík, Alžběta Šrůtková, and Irena Holubová. “Benchmark Engineering as a Design Instrument for Heterogeneous Information Systems”. In: *arXiv* (2026). Manuscript under second-round review. eprint: [2607.07175](https://arxiv.org/abs/2607.07175) (cs.DB). URL: <https://arxiv.org/abs/2607.07175>
14. Filip Ježek et al. *One Algebra to Query Them All: A Schema-Agnostic Foundation for Cross-Model Querying*. Manuscript under review. 2026. URL: <https://data.mmcatdb.com/papers/unified-algebra.pdf>
15. Jáchym Bártík, Filip Štrobl, and Irena Holubová. *Reducing the Cross-Model Tax: Query Optimization over Multi-Model Data*. Manuscript under review. 2026. URL: <https://data.mmcatdb.com/papers/query-opt.pdf>